

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF
ECONOMICS Fundamentals of Probability

Week 2 · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 academic
year



Week 1. We had the data. We described it: $\bar{x} = 4.605$, $s = 0.240$, the distribution is skewed to the left.

All of this is about the 9 numbers we have.

A common mistake But the real question is different. "What is the probability that next year's yield will be higher than 4.5?"

Descriptive statistics cannot answer this question. It requires the language of uncertainty.



Probability is not an end in itself. It is the grammar of statistical inference. The p -value, confidence interval, hypothesis—all are written in the language you will learn this week.

Learning objectives for week 2

By the end of this week, you will:

1. Construct an elementary event space for a random experiment and write the event as its subset.
2. State the Kolmogorov axioms and derive the main results from them.
3. Apply the rules of addition and multiplication.
4. Count the outcomes using combinatorics (permutation, combination).
5. Check the independence of two events and distinguish it from “non-coexistence”.
6. Check the probability with simulation.

Learning outcome CLO2. Apply the rules of probability, including conditional probability and Bayes' theorem, to economic problems (Bloom: Application).

Today's data set. movies-budget.csv — 2,467 films (2000–2018). We will work with 2,458 of them, with ratings.

We will study probability by calculating it in real data.

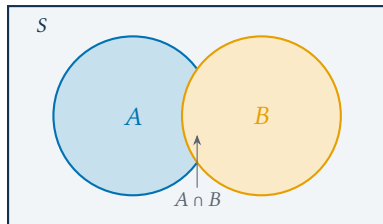
Elementary event space and events

Three basic concepts

Definition Random Experiment — a process whose outcome is not known in advance, but all possible outcomes are known.

Sample Space Ω — the set of all possible outcomes.

Event A — a subset of Ω : $A \subseteq \Omega$.



One die. $\Omega = \{1, 2, 3, 4, 5, 6\}$. “Even number” event: $A = \{2, 4, 6\}$. Two dice. $|\Omega| = 36$. “Total 7” event has 6 outcomes.

The dataset itself can be a space

Genre G PG PG-13 R Total

Action	0	6	178	174	358	Adventure	41	246	
	62	14	363	Comedy	3	39	266	292	600
Drama	5	106	339	479	929	Horror	0	3	51
	154	208							

Total	49	400	896	1	113	2	458
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Source: 2,467 films, 2000–2018; 9 are unrated

Experiment. We randomly select one film from the list.

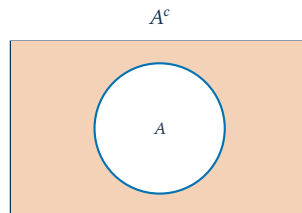
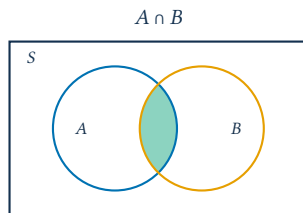
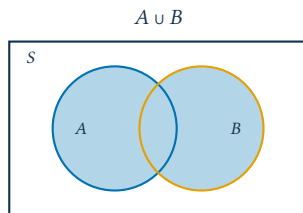
$n = 2,458$ films. Each is equally likely.

$X = \text{Drama}$ $P(X) = \frac{929}{2458}$ $X = \text{R}$
rating $P(X) = \frac{1}{113}$ $n(X) = 479$

$$P(X) = \frac{929}{2458} = 0,3779$$

Common Mistake Where did the 9 movies go? They were removed from space. This decision must be disclosed in the report—otherwise the denominators will be unclear.

Actions on Events



Action Readings in the Example of Movies

$A \cup B$ “ A or B ” (or both) Drama or R-rated: 1,563 movies

$A \cap B$ “ A and B ” Drama and R-rated: 479 movies A^c “ A not” Non-drama: 2,458
 $- 929 = 1,529$ movies

Axioms

Kolmogorov's axioms

Axioms (1933) Each event $\square \subseteq \square$ is associated with a number $P(\square)$ and:

A1 (non-negativity) $P(\square) \geq 0$

A2 (normalization) $P(\square) = 1$

A3 (additivity) If $\square \cap \square = \emptyset$,

$$P(\square \cup \square) = P(\square) + P(\square)$$

It follows from the axioms:

1. $P(\square) = 1 - P(\square)$

2. $P(\emptyset) = 0$

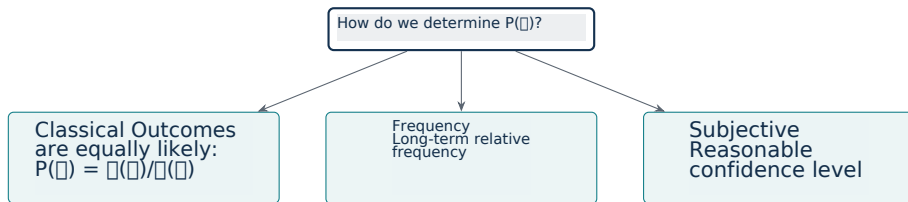
3. $0 \leq P(\square) \leq 1$

4. $\square \subseteq \square \Rightarrow P(\square) \leq P(\square)$

Proof of Result 1. $\square$ and $\square$ do not intersect, union is $\square$. By A3 and A2: $P(\square) + P(\square) = P(\square) = 1$. $\square$

Common Mistake Axioms do not tell us what probability is — they only tell us what rules it obeys. The next slide will decide where to get the value.

Where do we get the probability value?



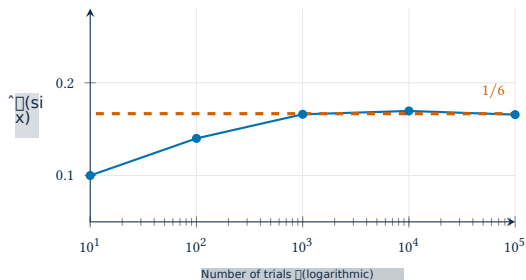
Approach When it works Economic example

Classical Symmetry based (dice, coin, random selection) Random one out of 2,458 movies

Frequency If the experiment can be repeated many times Credit default rate Subjective Non-repeating one-time event "Inflation will exceed 10% next year"

All three obey the same axioms. Therefore, the mathematics is the same — the interpretation is different.

Law of Large Numbers: Frequency Tends to Probability



Simulation, seed = 2026

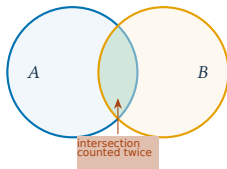
n	$\hat{p}$	$ \hat{p} - 1/6 $
10	0,100	0,0667
100	0,140	0,0267
1 000	0,166	0,0007
10 000	0,170	0,0029
100 000	0,166	0,0010

Common error Notice carefully. At $n = 1,000$, the error is smaller than at $n = 10,000$. The law of large numbers does not guarantee a monotonic decrease in error—it only guarantees convergence as $n \rightarrow \infty$. This becomes a precise formula in Week 7.

Addition rule

Theorem
Optional for $\square, \square$

$$P(\square \cup \square) = P(\square) + P(\square) - P(\square \cap \square)$$



In the example of movies. $\square$ = Drama, $\square$ = R rating.

$$P(\square) = 929/2458 = 0.3779 \quad P(\square) = 1113/2458 = 0.4528$$

$$P(\square \cap \square) = 479/2458 = 0.1949$$

$$P(\square \cup \square) = 0.3779 + 0.4528 - 0.1949 = 0.6359$$

$$\text{Check Count: } 929 + 1113 - 479 = 1563; 1563/2458 = 0.6359$$

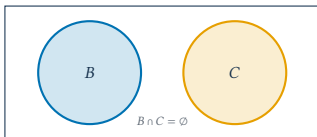
Common Error Without subtraction:
 $0.3779 + 0.4528 = 0.8308 - 0.1949$
more.

Mutually Exclusive Events

Definition Mutually Exclusive Events are events that cannot occur simultaneously: $A \cap B = \emptyset$.

In this case, $P(A \cap B) = 0$ and the addition rule is simplified:

$$P(A \cup B) = P(A) + P(B)$$



Example. A = R rating, B = G rating. A movie cannot have both ratings.

$$P(A \cup B) = \frac{1113}{2458} + \frac{49}{2458} = 0,4727$$

Audience Question A = Drama and B = R rating are not mutually exclusive?

No — 479 movies have both. Therefore, subtraction was necessary.

Combinatorics

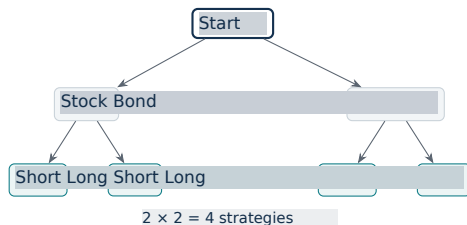
Why do we learn to count?

In the classical approach

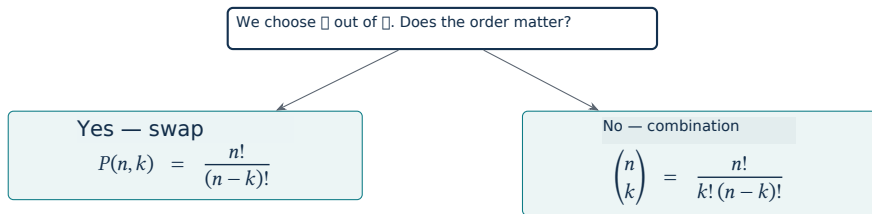
$$P(A) = \frac{n(A)}{n(S)}$$

So to find the probability, you need to count two numbers. If the space is large, you can't count by hand - formulas are needed.

Multiplication principle If step 1 can be done in n_1 way, and step 2 in n_2 way, then both can be done in $n_1 \times n_2$ way.



Swap or combination?



Situation Type Calculation

12 candidates, 3 people working group Combination $\binom{12}{3} = 220$
12 candidates, director, financier, marketer Swap $P(12, 3) = 1320$

20 stocks, 4 stocks, Combination $\binom{20}{4} = 4845$

Common mistake

$P(12, 3) = 1320$ and $\binom{12}{3} = 220$: the ratio is exactly $3! = 6$ — taking into account the order, each group is $k!$ times.

Combinatorics in probability: lottery

Problem. 6 numbers are randomly selected from 49. What is the probability of getting all 6?

Size of space — order not important:

$$n(S) = \binom{49}{6} = \frac{49!}{6!43!} = 13\,983\,816$$

There is only one winning combination: $\mathbb{P}(S) = 1$.

$$\mathbb{P}(S) = \frac{1}{13\,983\,816} \approx 7.15 \times 10^{-8}$$

That's one in 14 million — if you get one ticket every week, the average waiting time is $\approx 269\,000$ years.

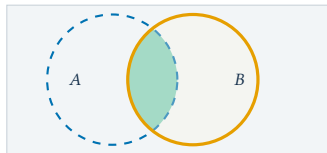
Audience question “I play the same numbers every week — they haven’t come up yet, now it’s time to come up.” What is the error in this reasoning called?

A common error is the gambler’s fallacy. The weights are independent — past outcomes do not affect future ones. The law of large numbers does not “restore balance,” it simply dilutes the past.

Conditional probability and the multiplication rule

Definition Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) > 0$$



new space = B

In the example of movies, A = Horror ($n(A) = 208$). How many horror movies are rated R? 154.

$$P(A|R) = \frac{154}{208} = 0,7404$$

Compare: $P(A) = 0.4528$.

B = Adventure ($n(B) = 363$), R-rated: 14.

$$P(B|R) = \frac{14}{363} = 0,0386$$

Information changes probability. Knowing the genre changes the probability of an R rating from 0.04 to 0.74.

Multiplication rule

Theorem We rewrite the definition of conditional probability:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

$$= P(A) \cdot P(A|B)$$

Verification (movies).

$$P(\text{Drama} \cap B) = P(\text{Drama}) \cdot P(B|\text{Drama})$$

$$= 0.3779 \times 0.5156 = 0.1949$$

✓ is the same as the directly calculated value.

Why is this rule important?
Often, $P(A \cap B)$ cannot be calculated directly, but $P(A)$ and $P(B|A)$ are known.

For example: the default of a customer on a loan is $P(\text{default}) \times P(\text{loss} | \text{default})$.

Independence

Independence: Three Equivalent Definitions

A definition $\square$ and $\square$ is said to be independent if one (and therefore all three) of the following holds:

$$(1) P(\square \cap \square) = P(\square) P(\square)$$

$$(2) P(\square | \square) = P(\square)$$

$$(3) P(\square | \square) = P(\square)$$

Meaning: Knowing $\square$ does not change our judgment of $\square$.

Let's check: Drama and R rating.

$$P(\square) P(\square) = 0.3779 \times 0.4528 = 0.1711$$

$$P(\square \cap \square) = 0.1949$$

$$0.1949 \neq 0.1711 \Rightarrow \text{related}$$

With the second form: $P(\square | \text{Drama}) = 0.5156$ vs $P(\square) = 0.4528$.

The ratio $0.1949/0.1711 = 1.14$ — Drama films are 14% more likely to receive an R rating.

Two Most Confused Concepts

Disjoint Disjoint

Definition $A \cap B = \emptyset$ $P(A \cap B) = P(A) P(B)$ $P(A \cap B) = 0 = P(A) P(B) > 0$ (if both > 0) $P(A|B) = 0 = P(A)$
Both cannot occur together Information about one does not change the other Example
The same movie cannot be rated G and R Flipping a coin twice

Common Misconception They are not opposites — they are almost opposites. If $P(A) > 0$, $P(B) > 0$, and $A \cap B = \emptyset$, then $P(A|B) = 0 \neq P(A)$ — so the disjoint events are maximally correlated.

Audience Question If you know that B happened, you know for sure that A did not. Does this mean that it is “not informative”?

Testing for independence in real data

Central Asian wheat yield, 135 observations

Median wheat yield by region = 1.997 t/ha. We define two events:

► A = "observation belongs to Uzbekistan" ► B = "productivity is above the median"

$$P(A) = 27/135 = 0.200 \quad P(B) = 67/135 = 0.496$$

$$P(A) P(B) = 0.0993$$

$$P(A \cap B) = 24/135 = 0.1778$$

$0.1778 \neq 0.0993 \Rightarrow$ the events are related.

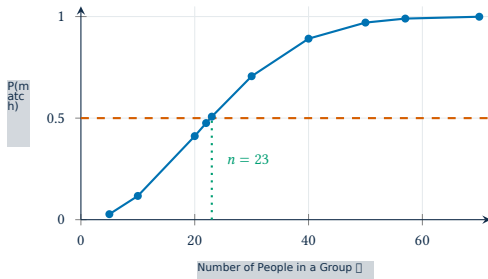
In other words:

$$P(B|A) = \frac{24}{27} = 0.889 \gg P(B) = 0.496$$

89% of Uzbekistan's years are above the regional median.

A common fallacy is that this statistical relationship is not a causal claim. Irrigated agriculture is dominant in Uzbekistan; rainfed agriculture is dominant in Kazakhstan. We will return to this distinction in week 13.

The Birthday Problem: When Intuition Lies



Solving by Complement. The probability that no one matches is:

$$P(\text{no matches}) = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{365 - n + 1}{365}$$

$$P(n) = 1 - P(\text{no matches})$$

n $P(n)$

10 0,117

23 **0,507**

30 0,706

50 0,970

Method. “At least one” problems are almost always easier to solve by complementing. This is a technique used throughout the course.

**Summar
y**

A common mistake is the prosecutor's fallacy — confusing $P(\text{evidence} \mid \text{not guilty})$ with $P(\text{evidence} \mid \text{guilty})$.

Our data also show the same difference:

$$P(\text{guilty} \mid \text{Horror}) = 0.740$$

$$P(\text{Horror} \mid \text{guilty}) = 0.138$$

Not the same two numbers — more than five times the difference.

Why is this an ethical issue? This confusion has led to wrongful convictions in real court cases. The result of statistical error is human fate.

The same mistake also occurs in medicine (screening tests) and credit scoring.

Week 3 Bayes' theorem relates these two conditional probabilities.

Common Student Mistakes and Misconceptions

Misconception Why it's wrong

“Events that are not mutually exclusive are dependent.” On the contrary — they are maximally dependent. If A occurs, B definitely does not occur: $P(A|B) = 0 \neq P(A)$.

“ $P(A \cup B) = P(A) + P(B)$.” This is only true when the intersection is empty. If we don't divide in the movies, the result is 0.8308 — instead of the actual 0.6359.

“ $P(A|B) = P(B|A)$.” $P(A|\text{Horror}) = 0.740$, but $P(\text{Horror} | A) = 0.138$. The denominators are completely different.

“The coin has landed heads 5 times — now it should land heads.” Gambler's error. The tosses are unrelated; the coin has no memory.

“The law of large numbers always minimizes error.” In our simulation, the error at $n=1000$ was smaller than at $n=10,000$. Convergence is guaranteed, not monotonicity.

“The probability can be greater than 1.” The A2 and monotonicity axioms forbid this. If your answer is > 1 , then the intersection is almost always counted twice.

“The combination and the permutation are almost the same.” $P(12, 3) = 1320$, $(12\ 3) = 220$ — a sixfold difference. Always ask: does the order matter?

Main conclusions

1. An event is a subset of a space. Solving a probability problem almost always begins with the correct definition of the space.

2. Three axioms hold the whole theory; the remaining rules are derived from them.

3. Don't forget to subtract the intersection in the addition rule — this is the most common calculation error.

4. “Does the order matter?” — the only important question in combinatorics.

5. Independence $\neq$ non-coexistence. Check for independence, don't assume.

6. Confusing $P(A|B)$ and $P(B|A)$ — the most costly mistake in statistics.

Next week Week 3 — Conditional probability and Bayes' theorem. Complete probability formula, tree diagrams, base rate error; diagnostic testing and credit scoring examples.

This week's assignments

Assignment Content Due Date

Required Reading Newbold Chapter 3; McClave Chapter 3; Heumann Chapter 6 (axioms) Until Lecture 2

Laboratory W02-lab — Monte Carlo Simulation and Empirical Probability In the laboratory lesson, a set of problems will be assigned to MT1; MT2 will be assigned to MT2: Week 3 Homework W02-homework — 6 problems, 40 points Week 3, Monday Project Phase 2 Source and context: 3 sources Week 3, Friday SI assignment Find the probability errors of SI Week 3, Friday

Common error in Week 3 Control work 1 — material from weeks 1-3. Write the problem and solve it: define the space, write the event, name the rule, then calculate.

1. Newbold, P., Carlson, W. & Thorne, B. Statistics for Business and Economics. Pearson. Chapter 3 — Probability.
2. McClave, J. T. & Sincich, T. (2018). Statistics, 13th ed. Pearson. Chapter 3.
3. Heumann, C., Schomaker, M. & Shalabh (2016). Introduction to Statistics and Data Analysis. Springer. Chapter 6 — Elements of Probability Theory.
4. Cetinkaya-Rundel, M. & Hardin, J. (2024). Introduction to Modern Statistics, 2nd ed. OpenIntro is a probability application.
5. DeGroot, M. & Schervish, M. (2012). Probability and Statistics, 4th ed. Chapter 1 (advanced).
6. Kolmogorov, A. N. (1933). Grundbegriffe der Wahrscheinlichkeitsrechnung — axiomatic basis.
7. Interactive: Seeing Theory, Brown University, chapters 1-2 — seeing-theory.brown.edu
8. Video: 3Blue1Brown, “Why probability of 0 does not mean impossible”; MIT OCW 18.05, lectures 1-3.